

Tharun Chaitanya Battula

Data Analyst

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Summary

Data Analyst with over four years of experience specializing in financial fraud detection and web traffic optimization for companies such as Stripe and Tech Mahindra. Achievements include improving fraud detection by 30% through automated data pipelines and enhancing decision-making efficiency by 40% with interactive dashboards. Promoted from Associate Data Analyst to Data Analyst within two years while collaborating cross-functionally to refine analytical models using advanced SQL and Python techniques.

Technical Skills

- Programming Languages: Python (Pandas, NumPy, Dask, Modin, SciPy, Statsmodels), SQL, R
- Data Analysis & Visualization: Statistical and Time-Series Analysis, Hypothesis Testing, Regression Models, Anomaly Detection
- ETL & Data Processing: AWS Glue, Apache Spark, PySpark, Airflow
- BI & Reporting Tools: Power BI, Tableau (Dashboard Creation, Data Validation, KPI Tracking), Excel (Advanced Formulas, Power Query, Solver)
- Machine Learning: Autoencoders, Isolation Forests, Clustering, Classification
- Big Data Tools: Hadoop, Hive, Apache Kafka
- Databases: SQL, MySQL, PostgreSQL, NoSQL (MongoDB), Snowflake
- Version Control: Git, GitHub
- Cloud & Data Storage: AWS (S3, EC2, EMR, MSK, Redshift)

Professional Experience

Data Analyst, Stripe, Inc.

01/2024 – Present | Remote, USA

- Enhanced financial fraud detection workflows through the integration of **AWS Glue and Apache Spark(PySpark)**, automating **data pipelines** from SQL databases and **S3 storage**; achieved a 30% increase in transaction risk analysis accuracy.
- **Collaborated with senior analysts** to execute SQL-based queries for transactional data analysis, uncovering hidden fraud patterns and improving risk assessment accuracy by 25%.
- Designed and optimized **relational database models** in **SQL Server**, developed complex **SQL queries** for fraud detection, and created dynamic **SSRS reports**, leading to improved transaction analysis and operational efficiency.
- Applied **wavelet transformations** for time-series anomaly detection, identifying irregular transaction behaviors to reduce undetected fraud risks. Improved detection speed by 35% and incorporated feedback from Agile team sprints to refine analysis and adjust detection parameters.
- Refined machine learning algorithms, including **Autoencoders and Isolation Forests**, achieving a 15% decrease in false positives while collaborating with cross-functional teams to enhance real-time anomaly detection capabilities for transaction patterns.
- Leveraged **Power BI (DAX, Power Query, data modeling)** to design interactive dashboards for anomaly detection, transaction risk analysis, and fraud trends. Implemented drillthroughs and custom filters, enhancing stakeholder decision-making speed by 40%.
- Implemented **AWS Data Governance Framework** by leveraging **AWS Lake Formation, AWS Glue Data Catalog, and IAM policies** to ensure secure, role-based access control and data integrity across business units.
- Optimized and managed **Apache Airflow** pipelines to orchestrate seamless data movement between **AWS S3 and Snowflake**, enhancing workflow efficiency, automating ETL processes, and ensuring high data reliability.

Data Analyst, Tech Mahindra

01/2022 – 12/2022 | India

- Executed data-driven stock market analysis within an **Agile framework**, partnering with **stakeholders to establish KPIs** and optimize **ETL workflows** on **AWS Glue**. Ensured real-time financial data ingestion, improving predictive trend accuracy by 30%.
- Revamped SQL query structures utilizing **Recursive CTEs and advanced Window Functions** to manage large-scale financial datasets; delivered adaptive time series aggregations leading to a 25% increase in data processing speed.
- Utilized **python libraries (Pandas, NumPy, Matplotlib, plotly, Sklearn)** for advanced data validation and cleaning, ensuring data integrity before model input.
- **Generated** recurring and **ad-hoc** reports on financial data using SQL and Python, **enhancing** trend analysis accuracy by 30%.
- Conducted **A/B testing** using **Python and statistical methodologies (t-tests, chi-square tests, Anova tests)** to evaluate investment strategy effectiveness, optimizing portfolio performance with data-driven decision-making.
- Implemented **Monte Carlo simulations** for risk assessment, integrating stochastic modeling and time-series decomposition to project potential stock price fluctuations, enabling portfolio managers to develop more robust investment strategies.
- Designed **Tableau dashboards** for data validation and anomaly detection, implementing automated statistical thresholding and **KPI variance tracking** to ensure financial data accuracy. Increased reporting efficiency by 40% for executive decision-making.

Associate Data Analyst, Cognizant

03/2020 – 12/2021 | India

- Conducted web traffic and conversion rate analysis by collaborating with cross-functional teams, gathering data from **Google Analytics and SQL databases**, ensuring data accuracy for further analysis and reporting, improving decision-making efficiency by 20%.
- Extracted, cleaned, and transformed large-scale data using **Advanced SQL (CTEs and Window Functions)**, optimizing data pipelines to improve query efficiency and reduce data processing time by 25%, ensuring accurate traffic segmentation.
- **Executed statistical analysis** on complex datasets using Python (Pandas, NumPy, Stats models, Plotly) to identify key factors affecting conversion rates, leveraging **hypothesis testing and regression models**, which **elevated** conversion insights and actionable recommendations by 30%.
- Developed interactive **Tableau dashboards** that tracked **15 key performance indicators**, enabling stakeholders to monitor real-time web traffic and conversion metrics; enhanced data visibility for informed business decision-making.
- Performed in-depth data validation through **Advanced Excel techniques (Macros, pivot tables)**, including **Power Query and Solver**, to automate reporting tasks; transformed insights into actionable strategies that increased overall conversion rate efficiency by 15%.

Education

Master of Science, New Jersey Institute of Technology

Newark, New Jersey, USA

Data Science

01/2023 – 05/2024

Projects

Wine Quality Prediction on AWS EMR (11/2023 – 12/2023)

- Accelerated model training speed twofold by harnessing PySpark functionalities and using AWS EMR cluster.
- Employed PySpark MLlib to build a **Random Forest Classifier model, Decision Tree Classifier**, and fine-tuning hyperparameters, achieving an improvement in model accuracy of 20%.
- Evaluated model performance metrics such as accuracy, F1 score, and Weighted Precision using Multiclass Classification Evaluator.

Certificates

- AWS Certified Data Engineer – Associate: Validation number - [615c7b36155f47d4afb6dcd59d1b5b72](#) [Expires on 2027-12-05] .
- Tableau Desktop Specialist: [Verification ID - 06af8fe363fa4626b1fb80824ee60cb4](#).
- Databricks Certified Associate Developer for Apache Spark 3.0 - Python.